

PEER-TO-PEER (P2P) DATASET FEDERATION

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About me

- Apprenticeship in Computer Systems and Networks
- Computer science and communication bachelor programme
- HES-SO Valais-Wallis (University of Applied Sciences and Arts Western Switzerland)
- Bachelor's work



HES-SO energypolis

P2P dataset federation

- PoC
- Not to move MRIQC web API
- Demonstrate that sharing IQMs between peers without a central server is feasible

Problem

- MRIQC IQMs – Montreal and Lausanne
- Analyze them together from a Jupyter notebook without any shared infrastructure
- Comparison:

	Central server (Web-API)	P2P
Best for	Public sharing	Closed consortiums, no infra available
Data	Uploaded to a third-party server	Stays at each institution
Setup	Simple for the user	More complex, but autonomous
Dependency	Server + funding	None

Who might find this interesting?

- Consortium of 4 hospitals wanting to compare IQMs
- A lab wanting to aggregate IQMs from its own distributed machines
- Situation where the central server is unreachable

What this thesis will concretely demonstrate

- A Jupyter notebook that opens and queries Parquet files distributed across 2-5 machines, as if they were local.